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# Week 1 : Assignment 1

**The due date for submitting this assignment has passed.**

**Due on 2026-02-04, 23:59 IST.**

## Assignment submitted on 2026-01-20, 00:16 IST

1) What is the primary goal of Information Retrieval (IR)? *1 point*

- To structure databases
- To seek out unstructured documents that satisfy a user's information need
- To perform numerical computations
- To design user interfaces

**Yes, the answer is correct.**
**Score: 1**
**Accepted Answers:**
*To seek out unstructured documents that satisfy a user's information need*

2) What does the term **stopword** refer to in IR? *1 point*

- Words that are rare in documents
- Words that are always capitalized
- Technical terms in a document
- Commonly used words that do not contribute significantly to content understanding

**Yes, the answer is correct.**
**Score: 1**
**Accepted Answers:**
*Commonly used words that do not contribute significantly to content understanding*

3) What is the main difference between controlled vocabulary and free-text vocabulary? *1 point*

- [ ] Controlled vocabulary uses human-assigned topics; free-text uses terms from the document
- [ ] Controlled vocabulary is cheaper to implement
- [ ] Free-text vocabulary is more ambiguous
- [x] Controlled vocabulary has predefined terms; free-text vocabulary is more flexible and allows any terms.

Partially Correct.
Score: 0.34
**Accepted Answers:**
*Controlled vocabulary uses human-assigned topics; free-text uses terms from the document*
*Free-text vocabulary is more ambiguous*
*Controlled vocabulary has predefined terms; free-text vocabulary is more flexible and allows any terms.*

4) In query optimization for Boolean retrieval, the best practice is to process terms: *1 point*

- In alphabetical order
- In order of increasing document frequency
- In order of decreasing document frequency
- Randomly

**Yes, the answer is correct.**
**Score: 1**
**Accepted Answers:**
*In order of increasing document frequency*

5) A key disadvantage of controlled vocabulary is: *1 point*

- Too many terms
- Cannot be used for indexing
- Expensive and time-consuming
- Cannot represent topic details

**Yes, the answer is correct.**
**Score: 1**
**Accepted Answers:**
*Expensive and time-consuming*

6) Ad hoc IR means: *1 point*

- Using SQL to retrieve documents
- Retrieval done with unlimited queries
- Retrieval evaluated using a finite set of queries
- No evaluation is needed

**Yes, the answer is correct.**
**Score: 1**
**Accepted Answers:**
*Retrieval evaluated using a finite set of queries*

7) What is a major drawback of using *grep*-like linear scanning for IR on large collections? *1 point*

- It cannot handle Boolean queries
- It is efficient for ranked retrieval
- It supports proximity queries well
- It requires scanning the entire collection for each query term

**Yes, the answer is correct.**
**Score: 1**
**Accepted Answers:**
*It requires scanning the entire collection for each query term*

8) Which of the following are stored in the dictionary of an inverted index? *1 point*

- [x] The actual term
- [x] Document frequency
- [ ] Position of term in each document

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*Controlled vocabulary uses human-assigned topics; free-text uses terms from the document*
*Free-text vocabulary is more ambiguous*
*Controlled vocabulary has predefined terms; free-text vocabulary is more flexible and allows any terms.*

4) In query optimization for Boolean retrieval, the best practice is to process terms: ***1 point***

- In alphabetical order
- (●) In order of increasing document frequency
- In order of decreasing document frequency
- Randomly

**Yes, the answer is correct.**
**Score: 1**
**Accepted Answers:**
*In order of increasing document frequency*

5) A key disadvantage of controlled vocabulary is: ***1 point***

- Too many terms
- Cannot be used for indexing
- (●) Expensive and time-consuming
- Cannot represent topic details

**Yes, the answer is correct.**
**Score: 1**
**Accepted Answers:**
*Expensive and time-consuming*

6) Ad hoc IR means: ***1 point***

- Using SQL to retrieve documents
- Retrieval done with unlimited queries
- (●) Retrieval evaluated using a finite set of queries
- No evaluation is needed

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- It supports proximity queries well
- (●) It requires scanning the entire collection for each query term

**Yes, the answer is correct.**
**Score: 1**
**Accepted Answers:**
*It requires scanning the entire collection for each query term*

8) Which of the following are stored in the dictionary of an inverted index? ***1 point***

- [x] The actual term
- [x] Document frequency
- [ ] Position of term in each document
- [x] Pointer to the start of the posting list

**Yes, the answer is correct.**
**Score: 1**
**Accepted Answers:**
*The actual term*
*Document frequency*
*Pointer to the start of the posting list*

9) You have two posting lists: ***1 point***

$T_1$ = [3, 10, 24, 40]
$T_2$ = [1, 10, 15, 24, 35, 40]

How many comparisons occur during a standard AND merge?

- 6
- (●) 7
- 8
- 9

**Yes, the answer is correct.**
**Score: 1**
**Accepted Answers:**
*7*

10) Let us assume two posting list sizes, respectively for two words "A" and "B" are |A| = 1200 and |B| = 300. Then what will be the size of A AND B (i.e. the number of documents containing both A and B)? ***1 point***

- 36
- 1200
- 300
- (●) 90

**No, the answer is incorrect.**
**Score: 0**
**Accepted Answers:**
*300*

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# Week 2 : Assignment 2

**The due date for submitting this assignment has passed.**

**Due on 2026-02-04, 23:59 IST.**

## Assignment submitted on 2026-02-02, 23:56 IST

1) What is document frequency? *1 point*

- ○ Number of times a term occurs in a document
- ◉ Number of documents containing the term
- ○ Total occurrences of a term in the corpus
- ○ Number of tokens in a document

**Yes, the answer is correct.**
**Score: 1**
**Accepted Answers:**
*Number of documents containing the term*

2) Posting lists may store: *1 point*

- [x] DocIDs
- [ ] Document lengths
- [x] Term frequencies
- [x] Term positions

**Yes, the answer is correct.**
**Score: 1**
**Accepted Answers:**
*DocIDs*
*Term frequencies*
*Term positions*

3) Zipf's law relates: *1 point*

- ○ Term frequency vs document length
- ○ Document rank vs popularity
- ○ Posting length vs dictionary size
- ◉ Term rank vs frequency

**Yes, the answer is correct.**
**Score: 1**
**Accepted Answers:**
*Term rank vs frequency*

4) Heaps' law predicts: *1 point*

- ◉ Vocabulary size as a function of corpus size
- ○ Posting list compression ratio
- ○ Query popularity distribution
- ○ Document frequency growth

**Yes, the answer is correct.**
**Score: 1**
**Accepted Answers:**
*Vocabulary size as a function of corpus size*

5) Which dictionary implementation supports prefix/wildcard queries efficiently? *1 point*

- ○ Hash-based dictionary
- ○ SPIMI dictionary
- ◉ Sort-based dictionary
- ○ Binary heap dictionary

**Yes, the answer is correct.**
**Score: 1**
**Accepted Answers:**
*Sort-based dictionary*

6) SPIMI does not require sorting because: *1 point*

- ○ It never stores docIDs
- ○ It merges lists during query time
- ○ It stores tokens unsorted permanently
- ◉ It keeps posting lists in hash tables directly

**Yes, the answer is correct.**
**Score: 1**
**Accepted Answers:**
*It keeps posting lists in hash tables directly*

7) Zipf's law implies that: *1 point*

- [ ] The vocabulary stops growing eventually
- [ ] Most words occur rarely
- [x] A few words dominate the corpus
- [ ] Word frequency follows a power law

Partially Correct.
Score: 0.33
**Accepted Answers:**
*Most words occur rarely*
*A few words dominate the corpus*
*Word frequency follows a power law*

8) Which scenario makes hash-based dictionaries unsuitable? *1 point*

- ○ Very large number of terms
- ◉ Need for range/prefix queries

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3) Zipf's law relates: *1 point*

- Term frequency vs document length
- Document rank vs popularity
- Posting length vs dictionary size
- (●) Term rank vs frequency

**Yes, the answer is correct.**
**Score: 1**
**Accepted Answers:**
*Term rank vs frequency*

4) Heaps' law predicts: *1 point*

- (●) Vocabulary size as a function of corpus size
- Posting list compression ratio
- Query popularity distribution
- Document frequency growth

**Yes, the answer is correct.**
**Score: 1**
**Accepted Answers:**
*Vocabulary size as a function of corpus size*

5) Which dictionary implementation supports prefix/wildcard queries efficiently? *1 point*

- Hash-based dictionary
- SPIMI dictionary
- (●) Sort-based dictionary
- Binary heap dictionary

**Yes, the answer is correct.**
**Score: 1**
**Accepted Answers:**
*Sort-based dictionary*

6) SPIMI does not require sorting because: *1 point*

- It never stores docIDs
- It merges lists during query time
- It stores tokens unsorted permanently
- (●) It keeps posting lists in hash tables directly

**Yes, the answer is correct.**
**Score: 1**
**Accepted Answers:**
*It keeps posting lists in hash tables directly*

7) Zipf's law implies that: *1 point*

- [ ] The vocabulary stops growing eventually
- [ ] Most words occur rarely
- [x] A few words dominate the corpus
- [ ] Word frequency follows a power law

**Partially Correct.**
**Score: 0.33**
**Accepted Answers:**
*Most words occur rarely*
*A few words dominate the corpus*
*Word frequency follows a power law*

8) Which scenario makes hash-based dictionaries unsuitable? *1 point*

- Very large number of terms
- (●) Need for range/prefix queries
- Need for dynamic updates
- Need for faster lookups

**Yes, the answer is correct.**
**Score: 1**
**Accepted Answers:**
*Need for range/prefix queries*

9) In the context of posting list compression, why is storing gaps between document IDs more efficient than storing the IDs themselves for frequent terms? *1 point*

- (●) Gaps are always smaller numbers, requiring fewer bits to encode.
- Gaps eliminate the need for a dictionary.
- Gaps allow for lossy compression of the index.
- Gaps make range queries faster.

**Yes, the answer is correct.**
**Score: 1**
**Accepted Answers:**
*Gaps are always smaller numbers, requiring fewer bits to encode.*

10) According to Zipf's Law with constant c=0.1, approximately what percentage of the text do the top 10 most frequent terms account for? *1 point*

- 10%
- 20%
- (●) 30%
- 40%

**Yes, the answer is correct.**
**Score: 1**
**Accepted Answers:**
*30%*

9:52 18%

# Week 3 : Assignment 3

**The due date for submitting this assignment has passed.**

**Due on 2026-02-11, 23:59 IST.**

## Assignment submitted on 2026-02-02, 23:59 IST

1) What is the role of an Analyzer during indexing? *1 point*

- It computes relevance scores
- It tokenizes and processes text
- It stores fields on disk
- It merges index segments

**Yes, the answer is correct.**
**Score: 1**
**Accepted Answers:**
*It tokenizes and processes text*

2) In Lucene, a Document is best described as: *1 point*

- A single term in the vocabulary
- A query representation
- A container for multiple fields
- A scoring function

**Yes, the answer is correct.**
**Score: 1**
**Accepted Answers:**
*A container for multiple fields*

3) When adding a **Field** to a **Document**, which of the following are legitimate concerns regarding how the field's value is handled? (Select all that apply.) *1 point*

- [x] Stored: Should the original value be saved so it can be retrieved later?
- [x] Indexed: Should the value be processed so it can be searched?
- [x] Tokenized: If indexed, should the text be broken into individual keywords?
- [x] Typed: Is the value a string, integer, or date?

**No, the answer is incorrect.**
**Score: 0**
**Accepted Answers:**
*Stored: Should the original value be saved so it can be retrieved later?*
*Indexed: Should the value be processed so it can be searched?*
*Tokenized: If indexed, should the text be broken into individual keywords?*

4) You are indexing a large corpus of documents. For the "CONTENT" field, you need to support phrase queries. Which **IndexOptions** setting provides the necessary data? *1 point*

- `TextField`
- `IndexOptions.DOCS`
- `IndexOptions.DOCS_AND_FREQS`
- `IndexOptions.DOCS_AND_FREQS _AND_POSITIONS`

**Yes, the answer is correct.**
**Score: 1**
**Accepted Answers:**
*`IndexOptions.DOCS_AND_FREQS _AND_POSITIONS`*

5) You want to add 1000 documents to a new PyLucene index. Which of the following are correct and necessary steps in the main indexing workflow? (Select all that apply) *1 point*

- [x] Use **lucene.initVM()** at the very start of your script.
- [ ] For each document, create a new **IndexWriter** instance.
- [x] Use **writer.addDocument(doc)** to add each document to the index.
- [x] Finally, close the **IndexWriter** with **writer.close()** to persist changes.

**Yes, the answer is correct.**
**Score: 1**
**Accepted Answers:**
*Use **lucene.initVM()** at the very start of your script.*
*Use **writer.addDocument(doc)** to add each document to the index.*
*Finally, close the **IndexWriter** with **writer.close()** to persist changes.*

6) Which of the following design choices help in efficient indexing? *1 point*

- [x] Not storing large text fields
- [x] Using StringField for categorical data
- [ ] Tokenizing numeric fields
- [x] Choosing appropriate analyzers

**Yes, the answer is correct.**
**Score: 1**
**Accepted Answers:**
*Not storing large text fields*
*Using StringField for categorical data*
*Choosing appropriate analyzers*

7) Which of the following statements about Lucene Documents are correct? *1 point*

- [x] A document can have multiple fields with the same name
- [ ] Field names must be unique per document
- [ ] Documents are immutable once indexed
- [ ] Documents directly store posting lists

Partially Correct.
Score: 0.5
**Accepted Answers:**
*A document can have multiple fields with the same name*

80

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*Tokenized: If indexed, should the text be broken into individual keywords?*

4) You are indexing a large corpus of documents. For the "CONTENT" field, you need to support phrase queries. Which **IndexOptions** setting provides the necessary data? ***1 point***

- ○ `TextField`
- ○ `IndexOptions.DOCS`
- ○ `IndexOptions.DOCS_AND_FREQS`
- ◉ `IndexOptions.DOCS_AND_FREQS _AND_POSITIONS`

**Yes, the answer is correct.**
**Score: 1**
**Accepted Answers:**
*`IndexOptions.DOCS_AND_FREQS _AND_POSITIONS`*

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- [x] Finally, close the **IndexWriter** with **writer.close()** to persist changes.

**Yes, the answer is correct.**
**Score: 1**
**Accepted Answers:**
*Use **lucene.initVM()** at the very start of your script.*
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**Accepted Answers:**
*Not storing large text fields*
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- [ ] Field names must be unique per document
- [ ] Documents are immutable once indexed
- [ ] Documents directly store posting lists

Partially Correct.
Score: 0.5
**Accepted Answers:**
*A document can have multiple fields with the same name*
*Documents are immutable once indexed*

8) Why is it usually inefficient to store large text fields in Lucene? ***1 point***

- ○ Stored fields affect term statistics
- ◉ Stored fields increase index size
- ○ Stored fields reduce recall
- ○ Stored fields cannot be compressed

**Yes, the answer is correct.**
**Score: 1**
**Accepted Answers:**
*Stored fields increase index size*

9) Which Lucene field type is most appropriate for document identifiers? ***1 point***

- ○ `TextField`
- ○ `StoredField`
- ◉ `StringField`
- ○ **IntPoint**

**Yes, the answer is correct.**
**Score: 1**
**Accepted Answers:**
*`StringField`*

10) You are indexing book metadata. For the "ISBN" field, you need exact lookups but will never display it in search results. For the "DESCRIPTION" field, you need full-text search and must display the original text. Which pair of field types is most appropriate? ***1 point***

- ○ ISBN: TextField (stored=False), Description: StringField
- ◉ ISBN: StringField (stored=False), Description: TextField (stored=True)
- ○ ISBN: StringField (stored=True), Description: StringField (stored=True)
- ○ ISBN: TextField (stored=True), Description: TextField (stored=False)

**Yes, the answer is correct.**
**Score: 1**
**Accepted Answers:**
*ISBN: StringField (stored=False), Description: TextField (stored=True)*

80

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# Week 4 : Assignment 4

**The due date for submitting this assignment has passed.**

**Due on 2026-02-18, 23:59 IST.**

## Assignment submitted on 2026-02-10, 15:54 IST

1) What is the main risk of using a *single analyzer* for indexing multilingual documents? *1 point*

- Increased index size
- Incorrect tokenization and stemming
- Slower query processing
- Inability to store fields

**Yes, the answer is correct.**
**Score: 1**
**Accepted Answers:**
*Incorrect tokenization and stemming*

2) What does storing a field enable? *1 point*

- Faster indexing
- Query-time analysis
- Retrieval of original field value
- Higher ranking score

**Yes, the answer is correct.**
**Score: 1**
**Accepted Answers:**
*Retrieval of original field value*

3) Why might indexing title, abstract, and body as separate fields improve retrieval? *1 point*

- Reduces index size
- Enables field-specific boosting
- Avoids stopword removal
- Eliminates need for analyzers

**Yes, the answer is correct.**
**Score: 1**
**Accepted Answers:**
*Enables field-specific boosting*

4) A document has a very long body field and a very short title field. Without field normalization, what is the likely effect? *1 point*

- Title dominates scoring
- Body dominates scoring
- Both contribute equally
- Document is ignored

**Yes, the answer is correct.**
**Score: 1**
**Accepted Answers:**
*Body dominates scoring*

5) How does maxDoc() differ from numDocs()? *1 point*

- maxDoc() excludes deleted docs
- numDocs() includes deleted docs
- maxDoc() includes deleted docs
- They are always equal

**Yes, the answer is correct.**
**Score: 1**
**Accepted Answers:**
*maxDoc() includes deleted docs*

6) Why might two indexes built from the same corpus appear different in LUKE? *1 point*

- [x] Different analyzers used
- [ ] Different field storage settings
- [ ] Different indexing time
- [ ] Different OS

**Partially Correct.**
**Score: 0.5**
**Accepted Answers:**
*Different analyzers used*
*Different field storage settings*

7) What additional information is needed in an index to handle **(information UW/3 science)**? *1 point*

- Term frequencies
- Document IDs
- Term positions
- Field mappings

**Yes, the answer is correct.**
**Score: 1**
**Accepted Answers:**
*Term positions*

8) What is a key limitation of basic Boolean retrieval (AND, OR, NOT)? *1 point*

- It cannot handle wildcards
- It does not support ranking
- It cannot process non-English text
- It requires too much memory

80

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Avoids stopword removal

Eliminates need for analyzers

**Yes, the answer is correct.**
**Score: 1**
**Accepted Answers:**
*Enables field-specific boosting*

4) A document has a very long body field and a very short title field. Without field normalization, what is the likely effect? ***1 point***

- Title dominates scoring
- (●) Body dominates scoring
- Both contribute equally
- Document is ignored

**Yes, the answer is correct.**
**Score: 1**
**Accepted Answers:**
*Body dominates scoring*

5) How does maxDoc() differ from numDocs()? ***1 point***

- maxDoc() excludes deleted docs
- numDocs() includes deleted docs
- (●) maxDoc() includes deleted docs
- They are always equal

**Yes, the answer is correct.**
**Score: 1**
**Accepted Answers:**
*maxDoc() includes deleted docs*

6) Why might two indexes built from the same corpus appear different in LUKE? ***1 point***

- [x] Different analyzers used
- [ ] Different field storage settings
- [ ] Different indexing time
- [ ] Different OS

**Partially Correct.**
**Score: 0.5**
**Accepted Answers:**
*Different analyzers used*
*Different field storage settings*

7) What additional information is needed in an index to handle **(information UW/3 science)**? ***1 point***

- Term frequencies
- Document IDs
- (●) Term positions
- Field mappings

**Yes, the answer is correct.**
**Score: 1**
**Accepted Answers:**
*Term positions*

8) What is a key limitation of basic Boolean retrieval (AND, OR, NOT)? ***1 point***

- It cannot handle wildcards
- (●) It does not support ranking
- It cannot process non-English text
- It requires too much memory

**Yes, the answer is correct.**
**Score: 1**
**Accepted Answers:**
*It does not support ranking*

9) What does the query **(information OW/2 science)** require for a match? ***1 point***

- "information" must appear exactly 2 words before "science"
- (●) "information" must appear within 2 words before "science"
- "information" and "science" must be in the same document
- "information" must appear after "science"

**Yes, the answer is correct.**
**Score: 1**
**Accepted Answers:**
*"information" must appear within 2 words before "science"*

10) If English and German documents are indexed using the same EnglishAnalyzer, what is the *most likely* consequence? ***1 point***

- German stopwords will be preserved
- (●) German words may be incorrectly stemmed or removed
- Queries will fail to parse
- Index will not build

**Yes, the answer is correct.**
**Score: 1**
**Accepted Answers:**
*German words may be incorrectly stemmed or removed*

80

9:53 17%

# Week 5 : Assignment 5

**The due date for submitting this assignment has passed.**

**Due on 2026-02-25, 23:59 IST.**

## Assignment submitted on 2026-02-18, 19:21 IST

1) Which of the following are limitations of Boolean retrieval? *1 point*

- [x] No ranking of results
- [x] Hard for non-expert users
- [x] Cannot handle free-text queries
- [x] Returns too many, or too few matched documents

**No, the answer is incorrect.**
**Score: 0**

**Accepted Answers:**
*No ranking of results*
*Hard for non-expert users*
*Returns too many, or too few matched documents*

2) Given: Document D terms: {apple, pie, recipe, sugar, flour}, Query Q = {apple, pie, recipe}. What is the Jaccard coefficient Jaccard(D, Q)? *1 point*

- (●) 0.6
- ( ) 0.67
- ( ) 0.75
- ( ) 1.0

**Yes, the answer is correct.**
**Score: 1**

**Accepted Answers:**
*0.6*

3) Which of the following factors are ignored by Jaccard similarity? *1 point*

- [x] Term frequency
- [x] Term importance
- [ ] Unique word overlap
- [x] Exact word ordering

**Yes, the answer is correct.**
**Score: 1**

**Accepted Answers:**
*Term frequency*
*Term importance*
*Exact word ordering*

4) Given: N = 1000 documents, df('term') = 10. Compute idf('term'). *1 point*

- ( ) 1
- (●) 2
- ( ) 3
- ( ) 4

**Yes, the answer is correct.**
**Score: 1**

**Accepted Answers:**
*2*

5) Which of the following best explains why inner product similarity is biased? *1 point*

- ( ) It ignores rare terms
- ( ) It favors documents with more unique terms
- (●) It favors longer documents
- ( ) It penalizes frequent terms

**Yes, the answer is correct.**
**Score: 1**

**Accepted Answers:**
*It favors longer documents*

6) Given two vectors: *1 point*
Q = (1, 1, 0)
D = (2, 2, 0)
What is cosine similarity(Q, D)?

- ( ) 0
- ( ) 0.5
- (●) 1
- ( ) 2

**Yes, the answer is correct.**
**Score: 1**

**Accepted Answers:**
*1*

7) Given a document d, term t appears 4 times in d, i.e. tf(t,d) = 4, N = 1000, df(t) = 100 in collection. Using TF weight = $1+\log_{10}(tf)$ and IDF = $\log_{10}(N/df)$, what is the length-normalized TF-IDF weight for t in d? *1 point*

- ( ) 0.008
- ( ) 1.92
- ( ) 0.81
- (●) 1.60

**Yes, the answer is correct.**
**Score: 1**

**Accepted Answers:**
*1.60*

9:53 17%

4) Given: N = 1000 documents, df('term') = 10. Compute idf('term'). *1 point*

- 1
- 2
- 3
- 4

**Yes, the answer is correct.**
**Score: 1**
**Accepted Answers:**
*2*

5) Which of the following best explains why inner product similarity is biased? *1 point*

- It ignores rare terms
- It favors documents with more unique terms
- It favors longer documents
- It penalizes frequent terms

**Yes, the answer is correct.**
**Score: 1**
**Accepted Answers:**
*It favors longer documents*

6) Given two vectors: *1 point*
Q = (1, 1, 0)
D = (2, 2, 0)
What is cosine similarity(Q, D)?

- 0
- 0.5
- 1
- 2

**Yes, the answer is correct.**
**Score: 1**
**Accepted Answers:**
*1*

7) Given a document d, term t appears 4 times in d, i.e. tf(t,d) = 4, N = 1000, df(t) = 100 in collection. Using TF weight = $1+\log_{10}(tf)$ and IDF = $\log_{10}(N/df)$, what is the length-normalized TF-IDF weight for t in d? *1 point*

- 0.008
- 1.92
- 0.81
- 1.60

**Yes, the answer is correct.**
**Score: 1**
**Accepted Answers:**
*1.60*

8) Given: N=1000, df("deep")=200, tf("deep", d)=10. Using TF weight = $(1+\log_{10}(tf))$ and IDF = $\log_{10}(N/df)$, what is the TF-IDF weight for "deep" in document d? *1 point*

- 0.34
- 1.39
- 1.98
- 1.11

**Yes, the answer is correct.**
**Score: 1**
**Accepted Answers:**
*1.39*

9) In VSM, why is the term independence assumption problematic? *1 point*

- It ignores semantic relationships between terms
- It overweights rare terms
- It requires too much computation
- It cannot handle long queries

**Yes, the answer is correct.**
**Score: 1**
**Accepted Answers:**
*It ignores semantic relationships between terms*

10) Compute the vector space similarity between the query **"digital cameras"** and the document **"digital cameras and video cameras"**. Assume total number of documents in the collection (N) is 10,000,000, df(**digital**)=10,000, df(**cameras**)=50,000, df(video)=100,000. *1 point*

Assume logarithmic term weighting for query and document, idf weighting for the query only (not document) and cosine normalization for the document only is applied for calculating the similarity score. Treat **and** as a stop word.

- 0.847
- 0.897
- 3.118
- 3.971

**Yes, the answer is correct.**
**Score: 1**
**Accepted Answers:**
*3.118*

80

9:54 17%

# Week 6 : Assignment 6

**The due date for submitting this assignment has passed.**

**Due on 2026-03-04, 23:59 IST.**

## Assignment submitted on 2026-02-24, 21:29 IST

1) In BM25, if b=0, what happens to document length normalization? *1 point*

- It is fully applied
- It is partially applied
- It is not applied
- It is inverted

**Yes, the answer is correct.**
**Score: 1**
**Accepted Answers:**
*It is not applied*

2) Why is IDF always finite for a term in a collection? *1 point*

- Because collection size is finite
- Because term frequency is always > 0
- Because log of zero is undefined
- Because df(t) ≥ 1 for terms in the collection

**Yes, the answer is correct.**
**Score: 1**
**Accepted Answers:**
*Because df(t) ≥ 1 for terms in the collection*

3) In a fielded retrieval model, increasing the weight of the TITLE field will: *1 point*

- Penalize long documents
- Increase the importance of matches in TITLE
- Reduce IDF values
- Eliminate BODY matches

**Yes, the answer is correct.**
**Score: 1**
**Accepted Answers:**
*Increase the importance of matches in TITLE*

4) A document has length 120 words, avgdl=100, b=0.75, k1=1.2, tf=8 for term "data". What is the BM25 TF component value? *1 point*

- ~1.88
- ~2.78
- ~3.45
- ~4.32

**No, the answer is incorrect.**
**Score: 0**
**Accepted Answers:**
*~1.88*

5) Which of the following are valid design choices in field-based retrieval? *1 point*

- [x] Using different weights for TITLE, ABSTRACT, BODY
- [ ] Using the same match function for all zones (mandatory)
- [x] Using different match functions for different fields
- [x] Ignoring BODY completely

**No, the answer is incorrect.**
**Score: 0**
**Accepted Answers:**
*Using different weights for TITLE, ABSTRACT, BODY*
*Using different match functions for different fields*

6) If N=1,000,000 and df('t')=1,000 documents, what is its IDF(t) (considering log base 10)? *1 point*

- 1
- 2
- 3
- 6

**Yes, the answer is correct.**
**Score: 1**
**Accepted Answers:**
*3*

7) In the SMART notation lnc.ltn, which components apply to the query? *1 point*

- [x] Log TF
- [x] IDF
- [x] Cosine normalization
- [ ] Binary TF

**No, the answer is incorrect.**
**Score: 0**
**Accepted Answers:**
*Log TF*
*IDF*

8) According to the Probability Ranking Principle (PRP), an optimal IR system should: *1 point*

- Rank documents by decreasing TF–IDF score
- Rank documents by decreasing probability of relevance
- Classify documents into relevant/non-relevant

9:54 17%

**Accepted Answers:**
*Increase the importance of matches in TITLE*

4) A document has length 120 words, avgdl=100, b=0.75, k1=1.2, tf=8 for term "data". What is the BM25 TF component value? ***1 point***

- ~1.88
- ~2.78
- ~3.45
- ~4.32

**No, the answer is incorrect.**
**Score: 0**
**Accepted Answers:**
*~1.88*

5) Which of the following are valid design choices in field-based retrieval? ***1 point***

- [x] Using different weights for TITLE, ABSTRACT, BODY
- [ ] Using the same match function for all zones (mandatory)
- [x] Using different match functions for different fields
- [x] Ignoring BODY completely

**No, the answer is incorrect.**
**Score: 0**
**Accepted Answers:**
*Using different weights for TITLE, ABSTRACT, BODY*
*Using different match functions for different fields*

6) If N=1,000,000 and df('t')=1,000 documents, what is its IDF(t) (considering log base 10)? ***1 point***

- 1
- 2
- 3
- 6

**Yes, the answer is correct.**
**Score: 1**
**Accepted Answers:**
*3*

7) In the SMART notation lnc.ltn, which components apply to the query? ***1 point***

- [x] Log TF
- [x] IDF
- [x] Cosine normalization
- [ ] Binary TF

**No, the answer is incorrect.**
**Score: 0**
**Accepted Answers:**
*Log TF*
*IDF*

8) According to the Probability Ranking Principle (PRP), an optimal IR system should: ***1 point***

- Rank documents by decreasing TF–IDF score
- Rank documents by decreasing probability of relevance
- Classify documents into relevant/non-relevant
- Use Boolean retrieval

**Yes, the answer is correct.**
**Score: 1**
**Accepted Answers:**
*Rank documents by decreasing probability of relevance*

9) The BIM weight ***1 point***

$$log\frac{N-df(i)}{df(i)}$$

is closely related to:

- TF
- Cosine similarity
- IDF
- Document length normalization

**Yes, the answer is correct.**
**Score: 1**
**Accepted Answers:**
*IDF*

10) For a query with two terms q1 and q2, where df(q1)=100, df(q2)=10,000, and N=1,000,000, what is the total BIM score for a document containing both terms? ***1 point***

- ~7.98
- ~5.99
- ~4.97
- ~3.98

**Yes, the answer is correct.**
**Score: 1**
**Accepted Answers:**
*~5.99*

80

9:54 17%

# Week 7 : Assignment 7

**The due date for submitting this assignment has passed.**

**Due on 2026-03-11, 23:59 IST.**

## Assignment submitted on 2026-03-05, 01:39 IST

1) What is a Language Model in the context of IR? *1 point*

- A model that translates between languages
- A probability distribution over sequences of words
- A model that checks grammar
- A neural network for text classification

**Yes, the answer is correct.**
**Score: 1**
**Accepted Answers:**
*A probability distribution over sequences of words*

2) What problem does smoothing solve in LMs? *1 point*

- Overfitting to training data
- Zero probability for unseen words
- Slow computation
- High memory usage

**Yes, the answer is correct.**
**Score: 1**
**Accepted Answers:**
*Zero probability for unseen words*

3) Which smoothing method uses a fixed λ for all documents? *1 point*

- Jelinek-Mercer smoothing
- Dirichlet smoothing
- Laplace smoothing
- Absolute discounting

**Yes, the answer is correct.**
**Score: 1**
**Accepted Answers:**
*Jelinek-Mercer smoothing*

4) Given a document with length |D| = 100, term "AI" occurs 5 times. What is P("AI"|D) using MLE? *1 point*

- 0.05
- 0.5
- 0.005
- 0.0005

**Yes, the answer is correct.**
**Score: 1**
**Accepted Answers:**
*0.05*

5) If a query term is missing from a document, without smoothing, P(Q|D) becomes: *1 point*

- 0
- 1
- 0.5
- Undefined

**Yes, the answer is correct.**
**Score: 1**
**Accepted Answers:**
*0*

6) λ = 0.7, P(t|D) = 0.2, P(t|C) = 0.02. Smoothed P(t|D) = ? *1 point*

- 0.066
- 0.074
- 0.168
- 0.146

**Yes, the answer is correct.**
**Score: 1**
**Accepted Answers:**
*0.146*

**Questions 7 – 10:**
Suppose we have a collection that consists of the 4 documents given in the below table.

| docID | Document text |
|---|---|
| 1 | click go the shears boys click click click |
| 2 | click click |
| 3 | metal here |
| 4 | metal shears click here |

Build a query likelihood language model for this document collection. Assume a mixture model between the documents and the collection, with both weighted at 0.5. Maximum likelihood estimation (MLE) is used to estimate both as unigram models.

7) What will be the ranking of the documents when the query is: 'shears'? *1 point*

- 1, 4, 2, 3
- 4, 1, 2, 3
- 3, 1, 2, 4
- 4, 2, 1, 3

**Yes, the answer is correct.**

80

9:54 17%

- (selected) 0.05
- 0.5
- 0.005
- 0.0005

**Yes, the answer is correct.**
**Score: 1**
**Accepted Answers:**
*0.05*

5) If a query term is missing from a document, without smoothing, P(Q|D) becomes: ***1 point***

- (selected) 0
- 1
- 0.5
- Undefined

**Yes, the answer is correct.**
**Score: 1**
**Accepted Answers:**
*0*

6) $\lambda = 0.7$, $P(t|D) = 0.2$, $P(t|C) = 0.02$. Smoothed $P(t|D) = ?$ ***1 point***

- 0.066
- 0.074
- 0.168
- (selected) 0.146

**Yes, the answer is correct.**
**Score: 1**
**Accepted Answers:**
*0.146*

**Questions 7 – 10:**
Suppose we have a collection that consists of the 4 documents given in the below table.

| docID | Document text |
|---|---|
| 1 | click go the shears boys click click click |
| 2 | click click |
| 3 | metal here |
| 4 | metal shears click here |

Build a query likelihood language model for this document collection. Assume a mixture model between the documents and the collection, with both weighted at 0.5. Maximum likelihood estimation (MLE) is used to estimate both as unigram models.

7) What will be the ranking of the documents when the query is: 'shears'? ***1 point***

- 1, 4, 2, 3
- (selected) 4, 1, 2, 3
- 3, 1, 2, 4
- 4, 2, 1, 3

**Yes, the answer is correct.**
**Score: 1**
**Accepted Answers:**
*4, 1, 2, 3*

8) What will be the ranking of the documents when the query is: 'click'? ***1 point***

- 2, 1, 3, 4
- 1, 4, 2, 3
- 2, 3, 4, 1
- (selected) 2, 1, 4, 3

**Yes, the answer is correct.**
**Score: 1**
**Accepted Answers:**
*2, 1, 4, 3*

9) What will be the ranking of the documents when the query is: 'click shears'? ***1 point***

- 4, 1, 2, 3
- 1, 4, 3, 2
- 4, 2, 1, 3
- (selected) 4, 1, 3, 2

**No, the answer is incorrect.**
**Score: 0**
**Accepted Answers:**
*4, 1, 2, 3*

10) What will be the score of document 3 for query 'click shears' under this smoothing? ***1 point***

- (selected) It becomes 0
- It is positive but small
- It is exactly 0.25
- It is exactly 0.5

**No, the answer is incorrect.**
**Score: 0**
**Accepted Answers:**
*It is positive but small*

80

9:54 17%

# Week 8 : Assignment 8

**The due date for submitting this assignment has passed.**

**Due on 2026-03-18, 23:59 IST.**

## Assignment submitted on 2026-03-17, 10:54 IST

1) What does KL Divergence measure between two probability distributions? *1 point*

- Their Euclidean distance
- Their cosine similarity
- Their information loss
- Their correlation coefficient

**Yes, the answer is correct.**
**Score: 1**
**Accepted Answers:**
*Their information loss*

2) Which of the following is a property of KL Divergence? *1 point*

- Symmetric
- Always equals 1
- Non-negative
- Follows triangle inequality

**Yes, the answer is correct.**
**Score: 1**
**Accepted Answers:**
*Non-negative*

3) Which Lucene query class is used for exact matching of a single term? *1 point*

- PhraseQuery
- TermQuery
- BooleanQuery
- WildcardQuery

**Yes, the answer is correct.**
**Score: 1**
**Accepted Answers:**
*TermQuery*

4) Which of the following are true about KL Divergence? *1 point*

- [ ] KLD(P||Q)=KLD(Q||P)
- [x] KLD(P||Q)≥0
- [x] KLD(P||Q)=0 if P=Q
- [ ] In order to get a ranked list, documents are arranged in the decreasing order of KLD(D||Query).

**Yes, the answer is correct.**
**Score: 1**
**Accepted Answers:**
*KLD(P||Q)≥0*
*KLD(P||Q)=0 if P=Q*

5) What is a key difference between **TermQuery** and **QueryParser** in Lucene? *1 point*

- TermQuery uses an analyzer, QueryParser does not
- QueryParser is case-sensitive, TermQuery is not
- TermQuery matches exact stored terms, QueryParser applies analysis
- QueryParser only works with StringField

**Yes, the answer is correct.**
**Score: 1**
**Accepted Answers:**
*TermQuery matches exact stored terms, QueryParser applies analysis*

6) In Jensen-Shannon Divergence, if JSD(P||Q)=0, what can be concluded? *1 point*

- P and Q are identical
- P and Q are orthogonal
- KLD(P||Q)=1
- KLD(Q||P)=0.5

**Yes, the answer is correct.**
**Score: 1**
**Accepted Answers:**
*P and Q are identical*

7) What is the range of Jensen-Shannon Divergence? *1 point*

- 0 to 1
- -1 to 1
- 0 to ∞
- Depends on the distributions

**Yes, the answer is correct.**
**Score: 1**
**Accepted Answers:**
*0 to 1*

8) In Lucene, if a field is indexed as **StringField**, which query method is more appropriate? *1 point*

- QueryParser
- TermQuery
- PhraseQuery
- FuzzyQuery

9:54 17%

Score: 1
**Accepted Answers:**
*TermQuery*

4) Which of the following are true about KL Divergence? *1 point*

- [ ] KLD(P||Q)=KLD(Q||P)
- [x] KLD(P||Q)≥0
- [x] KLD(P||Q)=0 if P=Q
- [ ] In order to get a ranked list, documents are arranged in the decreasing order of KLD(D||Query).

**Yes, the answer is correct.**
Score: 1
**Accepted Answers:**
*KLD(P||Q)≥0*
*KLD(P||Q)=0 if P=Q*

5) What is a key difference between **TermQuery** and **QueryParser** in Lucene? *1 point*

- ( ) TermQuery uses an analyzer, QueryParser does not
- ( ) QueryParser is case-sensitive, TermQuery is not
- (•) TermQuery matches exact stored terms, QueryParser applies analysis
- ( ) QueryParser only works with StringField

**Yes, the answer is correct.**
Score: 1
**Accepted Answers:**
*TermQuery matches exact stored terms, QueryParser applies analysis*

6) In Jensen-Shannon Divergence, if JSD(P||Q)=0, what can be concluded? *1 point*

- (•) P and Q are identical
- ( ) P and Q are orthogonal
- ( ) KLD(P||Q)=1
- ( ) KLD(Q||P)=0.5

**Yes, the answer is correct.**
Score: 1
**Accepted Answers:**
*P and Q are identical*

7) What is the range of Jensen-Shannon Divergence? *1 point*

- (•) 0 to 1
- ( ) -1 to 1
- ( ) 0 to ∞
- ( ) Depends on the distributions

**Yes, the answer is correct.**
Score: 1
**Accepted Answers:**
*0 to 1*

8) In Lucene, if a field is indexed as **StringField**, which query method is more appropriate? *1 point*

- ( ) QueryParser
- (•) TermQuery
- ( ) PhraseQuery
- ( ) FuzzyQuery

**Yes, the answer is correct.**
Score: 1
**Accepted Answers:**
*TermQuery*

9) What happens if you use a different analyzer during search than during indexing in Lucene? *1 point*

- ( ) No matches will be found
- ( ) The search will be slower
- (•) The results may be inaccurate or empty
- ( ) The index will be rebuilt

**Yes, the answer is correct.**
Score: 1
**Accepted Answers:**
*The results may be inaccurate or empty*

10) In the KL Divergence formula *1 point*

$$KLD(P||Q) = \sum_{x \in X} P(x) \log \frac{P(x)}{Q(x)}$$

What does a large value indicate?

- ( ) P and Q are similar
- (•) Using Q to approximate P results in high information loss
- ( ) P has higher entropy than Q
- ( ) Q is a good model for P

**Yes, the answer is correct.**
Score: 1
**Accepted Answers:**
*Using Q to approximate P results in high information loss*

9:55 16%

# Week 9 : Assignment 9

**The due date for submitting this assignment has passed.**

**Due on 2026-03-25, 23:59 IST.**

## Assignment submitted on 2026-03-17, 10:56 IST

1) Which of the following best describes Recall? *1 point*

- Fraction of retrieved documents that are relevant
- Fraction of relevant documents that are retrieved
- Fraction of documents classified correctly
- Harmonic mean of precision and recall

**Yes, the answer is correct.**
**Score: 1**
**Accepted Answers:**
*Fraction of relevant documents that are retrieved*

2) Which metric is most affected by class imbalance in IR? *1 point*

- Precision
- Recall
- Accuracy
- F1-score

**Yes, the answer is correct.**
**Score: 1**
**Accepted Answers:**
*Accuracy*

3) Which of the following are assumptions of the Cranfield paradigm? *1 point*

- [x] Relevance can be objectively determined
- [x] All relevant documents are known
- [ ] Relevance depends on document rank
- [x] All relevant documents contribute equally

**Yes, the answer is correct.**
**Score: 1**
**Accepted Answers:**
*Relevance can be objectively determined*
*All relevant documents are known*
*All relevant documents contribute equally*

4) Two systems have identical Precision, Recall, and F1 scores but one is better than the other. Which metric can distinguish them based on ranking quality? *1 point*

- Accuracy
- P@K
- R-Precision
- Average Precision

**Yes, the answer is correct.**
**Score: 1**
**Accepted Answers:**
*Average Precision*

5) Consider a collection that contains 10 relevant documents for a given query Q. An IR system retrieves a ranked list of 8 documents in response to query Q, for which the relevance assessments are given below. *1 point*

| Rank | Relevant? | Rank | Relevant? |
|---|---|---|---|
| 1 | YES | 5 | YES |
| 2 | NO | 6 | NO |
| 3 | NO | 7 | NO |
| 4 | NO | 8 | YES |

The non-interpolated average precision for this list is closest to:

- 0.18
- 0.25
- 0.36
- 0.60

**Yes, the answer is correct.**
**Score: 1**
**Accepted Answers:**
*0.18*

6) Consider a test collection in which, for a given query, documents are assigned gain values as follows: *1 point*

| Rating | Gain value |
|---|---|
| Not relevant | 0 |
| Marginally relevant | 1 |
| Relevant | 2 |
| Highly relevant | 3 |

For a particular query Q, a system retrieves documents D1, D2, D3, D4, D5 at ranks 1 through 5. Their gain values are 1, 1, 3, 0, 1 respectively. Calculate DCG at rank 5 with the discounting factor 1/log i (log base is 2).

- 3.51
- 3.98
- 4.32
- 5.04

**Yes, the answer is correct.**
**Score: 1**
**Accepted Answers:**

9:55 16%

5) Consider a collection that contains 10 relevant documents for a given query Q. An IR system retrieves a ranked list of 8 documents in response to query Q, for which the relevance assessments are given below. *1 point*

| Rank | Relevant? | Rank | Relevant? |
|---|---|---|---|
| 1 | YES | 5 | YES |
| 2 | NO | 6 | NO |
| 3 | NO | 7 | NO |
| 4 | NO | 8 | YES |

The non-interpolated average precision for this list is closest to:

- (●) 0.18
- ( ) 0.25
- ( ) 0.36
- ( ) 0.60

Yes, the answer is correct.
Score: 1
Accepted Answers:
*0.18*

6) Consider a test collection in which, for a given query, documents are assigned gain values as follows: *1 point*

| Rating | Gain value |
|---|---|
| Not relevant | 0 |
| Marginally relevant | 1 |
| Relevant | 2 |
| Highly relevant | 3 |

For a particular query Q, a system retrieves documents D1, D2, D3, D4, D5 at ranks 1 through 5. Their gain values are 1, 1, 3, 0, 1 respectively. Calculate DCG at rank 5 with the discounting factor 1/log i (log base is 2).

- ( ) 3.51
- ( ) 3.98
- (●) 4.32
- ( ) 5.04

Yes, the answer is correct.
Score: 1
Accepted Answers:
*4.32*

7) Consider the same gain values. For another query Q, a system retrieves documents D1, D2, D3, D4, D5 at ranks 1 through 5. Their gain values are 3, 2, 0, 3, 1 respectively. Calculate nDCG at rank 5 with the discounting factor 1/log i (log base is 2). *1 point*

- ( ) 0.71
- ( ) 0.80
- (●) 0.89
- ( ) 0.98

Yes, the answer is correct.
Score: 1
Accepted Answers:
*0.89*

8) Which evaluation scenarios are best matched with their metrics? *1 point*

- [x] Known-item search → MRR
- [x] Web search with graded relevance → nDCG
- [ ] Early-stage prototype → P@K
- [x] Legal e-discovery → MRR

No, the answer is incorrect.
Score: 0
Accepted Answers:
*Known-item search → MRR*
*Web search with graded relevance → nDCG*
*Early-stage prototype → P@K*

9) If the first relevant docs are retrieved at ranks: 3, 1, 5, 2 for 4 queries, what is the M *1 point*

- ( ) 0.46
- (●) 0.50
- ( ) 0.58
- ( ) 0.64

Yes, the answer is correct.
Score: 1
Accepted Answers:
*0.50*

10) If Cohen's κ = 0.72 between two assessors, their agreement is usually considered: *1 point*

- ( ) Poor
- ( ) Fair
- (●) Good
- ( ) Perfect

No, the answer is incorrect.
Score: 0
Accepted Answers:
*Fair*

80

9:56 16%

# Week 10 : Assignment 10

**The due date for submitting this assignment has passed.**

**Due on 2026-04-01, 23:59 IST.**

## Assignment submitted on 2026-03-28, 01:06 IST

1) In Relevance Model 1 (RM1) for query expansion, the expansion terms are derived from: *1 point*

- User click logs only
- Document frequency in the corpus
- Top retrieved documents weighted by query likelihood
- Pretrained neural embeddings

**Yes, the answer is correct.**
**Score: 1**
**Accepted Answers:**
*Top retrieved documents weighted by query likelihood*

2) Which of the following is an advantage of RM3 over RM1? *1 point*

- Reduces query drift by retaining original query terms
- Requires less computation
- Doesn't need smoothing
- Always improves recall

**Yes, the answer is correct.**
**Score: 1**
**Accepted Answers:**
*Reduces query drift by retaining original query terms*

3) In TREC Eval, what does the qrels file contain? *1 point*

- Ranked list of retrieved documents
- Query descriptions
- Ground truth relevance judgments
- Index statistics

**Yes, the answer is correct.**
**Score: 1**
**Accepted Answers:**
*Ground truth relevance judgments*

4) Which trec_eval option outputs **per-query results**? *1 point*

- `-m all_trec`
- `-q`
- `-m map`
- `-m P.10`

**Yes, the answer is correct.**
**Score: 1**
**Accepted Answers:**
*-q*

5) If a query has **10 relevant documents**, and the system retrieves **4 relevant documents in top 10**, what is **P@10**? *1 point*

- 0.10
- 0.25
- 0.40
- 0.60

**Yes, the answer is correct.**
**Score: 1**
**Accepted Answers:**
*0.40*

**Question 6 and 7:**

Given a query Q = {"machine", "learning"}, an initial retrieval is performed. After the initial retrieval, top 3 documents D1, D2, and D3 are retrieved with the following query likelihood scores: 1.76, 1.52, 1.44. Following are the smoothed term frequencies ($P(t|D_i)$) of some terms in each docs:

| Term | P(t\|D1) | P(t\|D2) | P(t\|D3) |
|---|---|---|---|
| machine | 0.2 | 0.1 | 0.3 |
| learning | 0.3 | 0.2 | 0.1 |
| algorithm | 0.1 | 0.4 | 0.2 |
| model | 0.4 | 0.3 | 0.4 |

6) What is $P_{RM1}$("algorithm"|R) in RM1 considering R is the latent relevance model? *1 point*

- 0.03
- 4.65
- 1.82
- 1.07

**Yes, the answer is correct.**
**Score: 1**
**Accepted Answers:**
*1.07*

7) If RM3 uses α=0.6 (weight to original query), and original query gives P("algorithm"|Q)=0, what is $P_{RM3}$("algorithm"|R)? *1 point*

- 7.28
- 4.82
- 4.28

80

9:56 16%

7) If RM3 uses α=0.6 (weight to original query), and original query gives P("algorithm"|Q)=0, what is $P_{RM3}$("algorithm"|R)? *1 point*

- 7.28
- 4.82
- 4.28
- 6.42

Yes, the answer is correct.
Score: 1

Accepted Answers:
*4.28*

For the rest of the questions (Q8 - Q15), you need to download a benchmark dataset, index it and run some retrieval experiments. We will be using the same FIRE dataset discussed in the lectures. Follow the steps. **Raise a question in the Discussion Forum in case you face any difficulty.**

1. Fill out this form and download the dataset (en.doc.2011-mx.tar): https://forms.gle/HLvkWpmGzwdPzwc46
2. Extract the downloaded .tar file. In the saved path, you will see two files:
   **(i) en_BDNews24.tgz** and
   **(ii) en_TheTelegraph_2001-2010.tgz.**
   These are also compressed files. Need to extract both the .tgz files.
3. You can extract them simply by right-clicking on them or using any unzipping software such as 7zip (https://www.7-zip.org/).
4. [optional] You may delete the two **.tgz** files.
5. Download the topics file containing 50 queries: https://www.isical.ac.in/~fire/data/topics/adhoc/en.topics.176-225.2012.txt
6. Download the qrel file: https://www.isical.ac.in/~fire/data/qrels/adhoc/en.qrels.176-225.2012-v2.txt.gz
   This is a .gz file. You can extract this file using your favourite zipping software.
7. Write code to make one index with all the documents stored in **en_BDNews24** and **en_TheTelegraph_2001-2010** folders. Please follow the instructions and codes (also shared in the Discussion Forum) discussed in the lecture.

Now answer the following questions:

8) How many relevant documents are there in total for the 50 queries, as indicated in the qrel file? *1 point*

- 2857
- 3076
- 3539
- 3607

No, the answer is incorrect.
Score: 0

Accepted Answers:
*3539*

9) Which query is having the highest number of relevant documents? *1 point*

- 204
- 189
- 192
- 224

No, the answer is incorrect.
Score: 0

Accepted Answers:
*204*

10) Which query is having the least number of relevant documents? *1 point*

- 208
- 187
- 194
- 223

No, the answer is incorrect.
Score: 0

Accepted Answers:
*187*

**Questions 11 - 15:** Write a retrieval framework (using Lucene/PyLucene) where you can change the retrieval model as well as parameters.

11) Perform retrieval using Language Model with Dirichlet Smoothing with MU set to 500 with the downloaded topics file. What is the mean average precision (MAP) after performing the retrieval with the topic set of 50 queries? Select the closest answer. *1 point*

- 0.30
- 0.28
- 0.36
- 0.40

Yes, the answer is correct.
Score: 1

Accepted Answers:
*0.36*

12) What is obtained precision at rank 5 (p@5) for the query number 200? Select the closest answer. *1 point*

- 0.1
- 0.2
- 0.3
- 0.4

No, the answer is incorrect.
Score: 0

Accepted Answers:
*0.4*

13) Perform retrieval using BM25 with k1 set to 1 and b set to 0.5 with the topics file. What is the overall obtained recall? Select the closest answer. *1 point*

- 0.6
- 0.7
- 0.8

80

9:56 16%

3607

No, the answer is incorrect.
Score: 0

Accepted Answers:
*3539*

9) Which query is having the highest number of relevant documents? *1 point*

- 204
- 189
- 192
- 224

No, the answer is incorrect.
Score: 0

Accepted Answers:
*204*

10) Which query is having the least number of relevant documents? *1 point*

- 208
- 187
- 194
- 223

No, the answer is incorrect.
Score: 0

Accepted Answers:
*187*

**Questions 11 - 15:** Write a retrieval framework (using Lucene/PyLucene) where you can change the retrieval model as well as parameters.

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- 0.30
- 0.28
- 0.36
- 0.40

Yes, the answer is correct.
Score: 1

Accepted Answers:
*0.36*

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- 0.1
- 0.2
- 0.3
- 0.4

No, the answer is incorrect.
Score: 0

Accepted Answers:
*0.4*

13) Perform retrieval using BM25 with k1 set to 1 and b set to 0.5 with the topics file. What is the overall obtained recall? Select the closest answer. *1 point*

- 0.6
- 0.7
- 0.8
- 0.9

Yes, the answer is correct.
Score: 1

Accepted Answers:
*0.8*

14) What is the reciprocal rank for the query number 186 after performing a retrieval using BM25 with k1 set to 1 and b set to 0.5 ? Select the closest answer. *1 point*

- 0.4
- 0.6
- 0.8
- 1.0

No, the answer is incorrect.
Score: 0

Accepted Answers:
*1.0*

15) What is the overall NDCG at rank 5 when a retrieval is performed using BM25 with k1 set to 1 and b set to 0.5? Select the closest answer. *1 point*

- 0.3
- 0.4
- 0.5
- 0.6

No, the answer is incorrect.
Score: 0

Accepted Answers:
*0.6*

80

9:56 16%

# Week 11 : Assignment 11

Your last recorded submission was on 2026-03-28, 01:11 IST

Due date: 2026-04-08, 23:59 IST.

1) In PageRank, what does the damping factor (d) typically represent? *1 point*

- ○ The probability of stopping the algorithm
- ◉ The probability of jumping to a random page
- ○ The number of inbound links
- ○ The number of outbound links

2) White Hat SEO refers to: *1 point*

- ○ Using unethical techniques to rank higher
- ◉ Following search engine guidelines and focusing on user experience
- ○ Buying links to improve ranking
- ○ Ignoring mobile optimization

3) In PageRank, an inbound link contributes more when it comes from: *1 point*

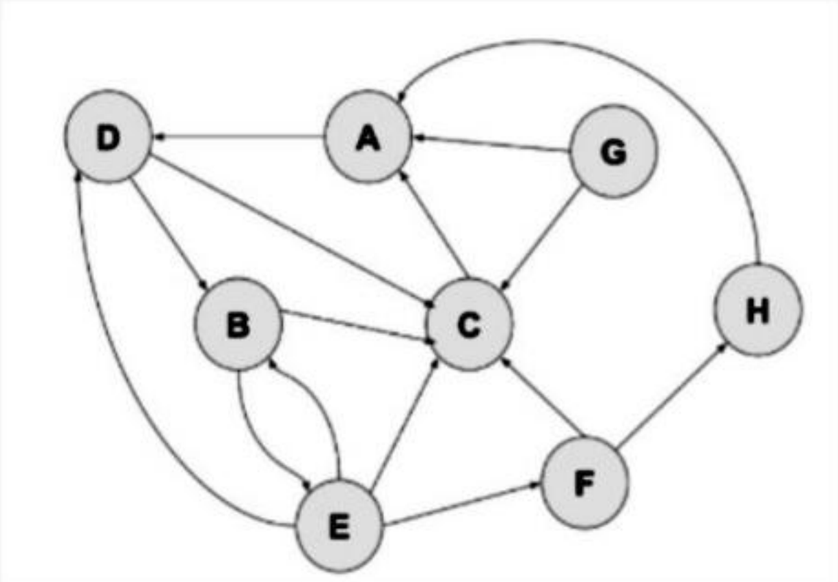
- ◉ A page with high PageRank
- ○ A page with many outbound links
- ○ A newly crawled page
- ○ A page with fewer terms

4) Given a PageRank damping factor d = 0.85, what fraction of rank mass comes from teleportation? *1 point*

- ◉ 0.15
- ○ 0.85
- ○ 1 / N
- ○ 0

5) Which matrix property is required for PageRank convergence? *1 point*

- ○ Symmetry
- ○ Diagonal dominance
- ◉ Stochasticity
- ○ Orthogonality




6) What will be the normalized authority score of node C in the above graph after first iteration? *1 point*

- ○ 1/12
- ○ 1/15
- ◉ 5/12
- ○ 5/15

7) What will be the normalized hub score of node E in the above graph after first iteration? *1 point*

- ○ 4/15
- ◉ 4/12
- ○ 1/12
- ○ 1/15

8) What will be the normalized authority score of node G in the above graph after second iteration? *1 point*

- ◉ 2/15
- ○ 4/15
- ○ 0
- ○ 2/12

9) What will be the normalized hub score of node A in the above graph after second iteration? *1 point*

- ○ 2/15
- ○ 1/15
- ◉ 2/45
- ○ 4/45

10) What will be the normalized authority score of node C in the above graph after second iteration?

- ◉ 12/35
- ○ 1/7

80

9:56 16%

2) White Hat SEO refers to: *1 point*

- Using unethical techniques to rank higher
- Following search engine guidelines and focusing on user experience
- Buying links to improve ranking
- Ignoring mobile optimization

3) In PageRank, an inbound link contributes more when it comes from: *1 point*

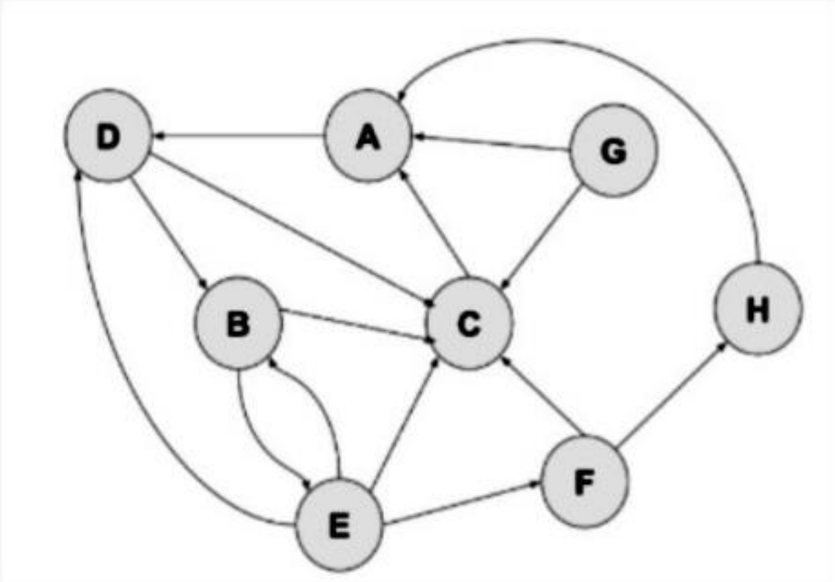
- A page with high PageRank
- A page with many outbound links
- A newly crawled page
- A page with fewer terms

4) Given a PageRank damping factor d = 0.85, what fraction of rank mass comes from teleportation? *1 point*

- 0.15
- 0.85
- 1 / N
- 0

5) Which matrix property is required for PageRank convergence? *1 point*

- Symmetry
- Diagonal dominance
- Stochasticity
- Orthogonality




6) What will be the normalized authority score of node C in the above graph after first iteration? *1 point*

- 1/12
- 1/15
- 5/12
- 5/15

7) What will be the normalized hub score of node E in the above graph after first iteration? *1 point*

- 4/15
- 4/12
- 1/12
- 1/15

8) What will be the normalized authority score of node G in the above graph after second iteration? *1 point*

- 2/15
- 4/15
- 0
- 2/12

9) What will be the normalized hub score of node A in the above graph after second iteration? *1 point*

- 2/15
- 1/15
- 2/45
- 4/45

10) What will be the normalized authority score of node C in the above graph after second iteration? *1 point*

- 12/35
- 1/7
- 4/35
- 1/15

You may submit any number of times before the due date. The final submission will be considered for grading.

Submit Answers

80

9:57 16%

Week 12 Feedback Form : Introduction to Information Retrieval!!

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# Week 12 : Assignment 12

Your last recorded submission was on 2026-04-08, 21:02 IST

Due date: 2026-04-15, 23:59 IST.

1) Which of the following best defines Learning to Rank (LTR)? *1 point*

- ○ A rule-based ranking framework
- ○ A probabilistic retrieval model
- ◉ A machine learning approach to learning ranking functions from data
- ○ A query expansion technique

2) Which of the following is a key innovation of RoBERTa over BERT? *1 point*

- ○ Adding more attention heads
- ◉ Removing Next Sentence Prediction
- ○ Using CNN layers
- ○ Adding reinforcement learning

3) If a term-document matrix has dimensions 1000×500 and rank 200, and we choose k=50 for SVD approximation, what is the size of Uk? *1 point*

- ◉ 1000×50
- ○ 50×500
- ○ 1000×200
- ○ 200×50

4) Which of the following are true about Sentence-BERT? *1 point*

- ☑ It encodes queries and documents separately
- ☐ It uses cross-encoder architecture
- ☐ It is designed for efficient semantic search
- ☐ It processes query and document together

5) LambdaRank improves RankNet by: *1 point*

- ○ Using gradient boosting
- ◉ Scaling gradients by NDCG gain
- ○ Adding more hidden layers
- ○ Using listwise loss

6) Why do we often assign lower weights to expansion terms than original query terms? *1 point*

- ○ To save memory
- ◉ To reduce influence of less certain terms on ranking
- ○ Because expansion terms are always irrelevant
- ○ To avoid computing embeddings

7) In LSI, the reduced-rank approximation Ak minimizes which quantity? *1 point*

- ○ KL-divergence
- ○ Cosine similarity loss
- ◉ Frobenius norm of A−Ak
- ○ Determinant of covariance matrix

8) Which combination best reflects a practical IR system using LSI? *1 point*

- ○ Replace original scores entirely with LSI scores
- ○ Use LSI only for query expansion
- ◉ Linearly combine original scores with LSI scores
- ○ Use LSI only for indexing

9) Key limitations of static embeddings include: *1 point*

- ☑ Same vector for polysemous words
- ☑ Inability to capture context
- ☐ High dimensionality
- ☐ Slow training time

10) Which embedding method is most likely to capture phrase-level semantics for query expansion? *1 point*

- ○ Word2Vec (Skip-gram/ CBoW)
- ○ GloVe
- ○ FastText
- ◉ BERT sentence embeddings

You may submit any number of times before the due date. The final submission will be considered for grading.

Submit Answers

80